Yabetse Tesfaye Alemu

Software Engineer

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EDUCATION

Software Engineering 07/2014 – 07/2019
American College of Technology Addis Ababa, Ethiopia
Relevant Coursework:
- Data Structures & Algorithms
- Web & Mobile Application Development
- Machine Learning & Artificial Intelligence
- Distributed Systems & Cloud Computing
- Operating Systems & System Programming
- Computer Networking & Network Security
- Database Systems & Data Modeling
- Software Engineering & Design Patterns
- Parallel & Concurrent Programming
- Computer Architecture

Coding Academy 10/2022 – 10/2023
A2SV Addis Ababa, Ethiopia
Successfully completed a rigorous boot camp focused on data structures and algorithms, built to prepare participants for software engineering interviews at leading tech firms.

PROFESSIONAL EXPERIENCE

Full Stack Developer (Internal Tooling) 2020 – 2021
Esklate
- Built an automation platform with Slack integration that streamlined Google Sheets workflows, adopted by over 200 employees across four time zones and improving operational efficiency.
- Designed a timezone-aware notification system using Redis and Celery job queues, reducing missed deadlines by 30% through personalized alerts.
- Developed a web-based dashboard for real-time project tracking and performance insights, enabling leadership to make data-driven decisions.
- Implemented a Kubernetes-based orchestration system that improved scalability and reduced deployment times by 60%.
- Engineered highly available Go microservices for container monitoring and management, helping maintain 99.9% system uptime.

React Frontend Developer 2021 – 2023
Kuraz Tech Engineering PLC
- Collaborated within a 30+ member frontend team to build responsive, high-performance interfaces, improving application load times by 40% through code optimization and lazy loading.
- Contributed to the migration to Next.js with SSR/SSG, improving SEO performance and increasing user engagement by 25%.
- Mentored junior developers through code reviews and pair programming, reducing onboarding time by 30%.
- Translated UX wireframes into reusable React components, helping improve overall user satisfaction by 15%.

Backend Engineer at Afro-Chat 2023 – 2026
Esklate
- Engineered backend and ML infrastructure across three major projects: Bazar, GAN Observatory, and RustV OS Lab
- Built the intelligence layer of Bazar, an AI-driven marketplace for homes and cars, delivering personalized listings through a TensorFlow and scikit-learn recommendation engine fed by an automated Python and Selenium data collection pipeline.

- Developed real-time ML serving infrastructure for GAN Observatory, streaming live training telemetry to browsers at 20 fps over FastAPI WebSockets, with thread-safe mid-run controls, a SQLite experiment registry, and on-demand model serving endpoints.
- Demonstrated deep systems expertise through RustV OS Lab, a bare metal RISC-V kernel in Rust covering boot, virtual memory, preemptive scheduling, user mode syscalls, and a hand-built DMA block driver with a persistent filesystem.
- Directed the migration from Flask + MongoDB to an async FastAPI + SQLAlchemy architecture, applying Domain-Driven Design and Test-Driven Development, which reduced API latency by 35%.
- Streamlined CI/CD workflows with GitHub Actions, Dockerized deployments, and automated testing, shortening release cycles by 20%.
- Built Python-based automation tools that cut container build times by 40%, improving development efficiency.
- Contributed to open-source Linux projects, improving developer documentation and system tooling.

PROJECTS

Adwa-Based Game

Mobile Role-Playing Game (Unity)

- Developed an immersive mobile RPG combining engaging gameplay with educational storytelling to portray Ethiopia's historic victory at the Battle of Adwa.
- Enhanced narrative depth through scripted and animated cinematic cutscenes depicting key historical moments such as the Wuchalé Treaty and Bahta Hagos' martyrdom.
- Implemented core gameplay systems including scene transitions, event triggers, and character movement to ensure smooth player interaction.
- Contributed to the modeling and design of 3D characters, shaping the protagonist and historical figures with period-accurate visual details.
- Collaborated on level design and UI integration to create intuitive controls and dynamic environments that improved player engagement.

Tech stacks: Unity, C#, 3D Modeling, Animation & Cinematics, Level Design, Mobile Game Development, UI Integration

AI-RAD

Healthcare Screening & Records Platform

- Developed a secure web application that enabled hospitals to perform COVID-19 screenings, digitize patient records, and manage data through a centralized system, improving accessibility and operational efficiency.
- Built role-based access control for doctors, nurses, and administrators, ensuring patient data stayed secure, organized, and instantly retrievable across departments.
- Designed screening workflows with symptom-based patient intake forms, enabling faster triage and reducing manual paperwork for hospital staff.
- Recognized with 1st place in the 2020 UNDP & Ministry of Innovation and Technology (MINT) COVID-19 Innovation Challenge, securing a 580,000 ETB grant for the solution.

Tech stacks: React, JavaScript, Node.js, Firebase Authentication, Firestore, REST APIs, Role-Based Access Control, Responsive Web Design

RustV OS Lab 🔗

Bare-Metal RISC-V Operating System

- Engineered a no_std RISC-V kernel in Rust that boots directly on QEMU without BIOS or SBI firmware, with a custom assembly entry point, stack initialization, boot-hart selection, and machine-mode runtime.
- Implemented trap handling, Sv39 virtual memory, page tables, physical memory protection, system calls, and user-mode fault isolation using full register context frames. • Built a preemptive round-robin scheduler driven by CLINT timer interrupts, with assembly context switching and support for concurrent user-mode tasks.
- Developed a virtio-MMIO block driver from scratch with modern and legacy transport support, split virtqueues, DMA sector operations, feature negotiation, and persistent storage.
- Created TinyFS and an interactive UART shell for persistent file operations, device inspection, and raw-sector access, validated through automated host tests and end-to-end QEMU integration tests.

Tech stacks: Rust, RISC-V Assembly, QEMU, Sv39 Virtual Memory, Virtio-MMIO, DMA, TinyFS, UART, Cargo, GNU Make, GitHub Actions

GAN Observatory 🔗

Real-Time GAN Training Observatory

- Engineered a real-time GAN observatory that streams training telemetry to an interactive browser dashboard at approximately 20 FPS using FastAPI, WebSockets, and a background training engine.
- Implemented a dependency-light NumPy GAN from scratch with manual forward propagation, backpropagation, and Adam optimization, supporting live hyperparameter updates without restarting training.
- Built interactive visualizations for generated distributions, discriminator decision boundaries, learned feature spaces, RBF-MMD, precision, recall, and automatic mode-collapse detection.
- Developed reproducible experiment tracking and model serving with SQLite, allowing users to save configurations, seeds, metric histories, and generator weights, then serve new samples through REST APIs.
- Containerized the platform with Docker and established automated quality gates using GitHub Actions, pytest, and Ruff, while retaining TensorFlow/Keras support for larger MLP and DCGAN experiments.

Tech stacks: Python, NumPy, TensorFlow/Keras, FastAPI, WebSockets, SQLite, JavaScript, Docker, GitHub Actions

ImmuStore DB 🔗

Distributed Database Engine Built from Scratch

- Engineered a distributed database engine from scratch in pure Python, featuring a copy-on-write B+ tree storage core with O(log n) reads, writes, sorted scans, lazy record loading, and append-only persistence with no third-party runtime dependencies.
- Implemented crash-safe durability and MVCC using shadow-paging commits, CRC32 record validation, double-buffered metadata blocks, torn-write recovery, lock-free point-in-time snapshots, and optimistic transactions with write-conflict detection.
- Built a Redis-compatible asynchronous network server using RESP and asyncio, supporting standard Redis clients, real-time prefix-based change data capture, concurrent connections, and live set/delete event streaming.
- Developed distributed replication through a from-scratch Raft consensus implementation with leader election, replicated logs, automatic failover, TCP cluster communication, and exactly-once application of committed operations.
- Added document collections, secondary indexes, query planning, TTL expiration, Prometheus metrics, administrative health endpoints, Docker deployment, and deterministic fault-injection testing across crashes, network partitions, corrupted metadata, and torn disk writes.

Tech stacks: Python, asyncio, B+ Trees, MVCC, Raft Consensus, Redis RESP, TCP Networking, SQLite, Prometheus, Docker, Docker Compose, pytest, Ruff, GitHub Actions

Bazar

AI-Driven Marketplace for Homes and Cars

- Designed and implemented an AI-powered recommendation engine using TensorFlow and scikit-learn to analyze user behavior and deliver highly personalized listings.
- Built a Selenium-based web scraper to automatically aggregate large volumes of real-time listings from multiple sources, maintaining a rich and up-to-date dataset.
- Improved recommendation accuracy by over 40% using clustering techniques and collaborative filtering to better match users with relevant listings.
- Developed a cross-platform Flutter mobile application using Test-Driven Development, ensuring scalability, maintainability, and a smooth user experience.

Tech stacks: Python, TensorFlow, scikit-learn, Selenium, Flutter, Dart, Collaborative Filtering, Clustering, Test-Driven Development

ገበጣ Gebeta 🔗

Traditional Ethiopian Mancala Game with AI

- Developed a culturally themed Gebeta game for desktop and web using Java Swing and TypeScript, supporting local multiplayer, three AI difficulty levels, responsive controls, animations, and sound effects.
- Built a time-bounded AI opponent using iterative deepening, minimax, alpha-beta pruning, transposition tables, principal-move ordering, and tactical capture prioritization.
- Designed a UI-independent rules engine covering stone sowing, captures, extra turns, illegal moves, scoring, and end-game collection, enabling reliable testing and AI simulations.
- Implemented versioned game persistence for saved sessions and player statistics using Java Preferences and browser localStorage, with defensive corruption recovery.
- Automated unit testing, cross-platform builds, GitHub Pages deployment, and native macOS packaging through Gradle, Vite, GitHub Actions, and jpackage.

Tech stacks: Java, Java Swing, TypeScript, HTML, CSS, Minimax, Alpha-Beta Pruning, Web Workers, Gradle, Vite, JUnit 5, GitHub Actions, GitHub Pages.

Play in Browser -> https://yabtesfu.github.io/Gebeta-Game/

Hulu Code
Code Executor Web Application
- Built an interactive, multi-language code execution platform supporting 7+ programming languages, frameworks, and databases, including Python, JavaScript, and SQL.
- Designed an intuitive interface for writing, running, and sharing code in real time, improving collaboration among developers.
- Integrated cloud-based execution to enable fast experimentation, significantly reducing environment setup time for testing and debugging.

Tech stacks: React, Next.js, JavaScript, Monaco Editor, Node.js, REST APIs, Cloud Code Execution (Judge0), SQL, Vercel

SKILLS

Programming Languages: Python, Rust, Go, Java, C#, TypeScript, JavaScript, Dart, SQL, RISC-V Assembly, HTML5, CSS3

Backend & Networking: FastAPI, Node.js, SQLAlchemy, REST APIs, WebSockets, asyncio, Celery, Redis, TCP Networking, Redis RESP, Firebase Authentication, Firestore

AI, Machine Learning & Algorithms: TensorFlow/Keras, scikit-learn, NumPy, GANs, Collaborative Filtering & Clustering, Minimax, Alpha-Beta Pruning, Iterative Deepening, Model Evaluation, Data Processing Pipelines (Selenium), LLM Prompt Design

Storage, Observability & Deployment: SQLite, MongoDB, Persistent Storage, Prometheus Metrics, Kubernetes, GitHub Pages, Vercel, Native Application Packaging, AWS/GCP Deployment Fundamentals

Systems & Database Engineering: Operating Systems, Distributed Systems, B+ Trees, MVCC, Raft Consensus, Virtual Memory, Preemptive Scheduling, Concurrency, File Systems, Crash Recovery

Frontend & Application Development: React, Next.js, Flutter, Unity, Java Swing, Tailwind CSS, Responsive Web Design, Web Workers, Interactive Data Visualization

DevOps, Testing & Tooling: Docker, Docker Compose, Kubernetes, Git, GitHub Actions, CI/CD, pytest, JUnit 5, Ruff, Gradle, Vite, QEMU, GNU Make, Test-Driven